

Probability: A measures of chance

In every terminology, probability can be thought of as a measure of the chance or likelihood that a particular outcome will occur. This occurrence is measured by numerical statements.

Probability and possibility

Sometimes a distinction is made between probability and possibility. Possibility precedes probability. We cannot make intelligent prediction unless we know at least what is possible. We can hardly predict who is going to win next world cup football unless we know the possible participants. Possibility thus refers to exhaustiveness of outcomes, while probability refers to their predictive behavior.

Set and operations

Well organized collection of elements is called set.

$$A = \{a, b, c, d\}$$

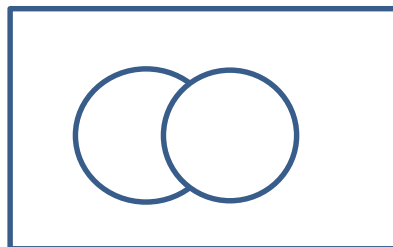
Universal set:

An universal set is the set of all elements that may possibly be considered in a particular discussion.

$$U = \{x | x \text{ is total number of spots uppermost on the two dice}\}$$

Venn diagram:

A helpful scheme to illustrate the relationship between sets and set operations.



Empty set:

Any set that has no element in it is called an empty, or null set, designated with $\emptyset$.

Subset:

A set A is a subset of the set B if and only if for each $x \in A$, then $x \in B$.

The set $A = \{1, 2, 3\}$ is a subset of the set $B = \{1, 2, 3, 4, 5, 6, 7\}$

Proper subset:

If there is at least one element of B that is not in A , then A is a proper subset of B , written as $A \subset B$

The set $A = \{1, 2, 3\}$ is a subset of the set $B = \{1, 2, 3, 4, 5, 6, 7\}$

Then $A \subset B$

Equal set:

Two sets A and B are said to be equal, or $A = B$, if and only if $A \subseteq B$ and $B \subseteq A$.

Unit set:

If a set contain only one element, it is called unit set or singleton set.

Finite and infinite sets:

A set is called finite if its elements are equal in number to some specifiable nonnegative integers. It is called infinite if the number of its elements is greater than any positive integer.

Operation with sets**Intersection:**

Given the sets A and B, which are subsets of U, the intersection of A and B written as $A \cap B$, is the set of all elements that are members of both sets.

For example

$$U = \{x | x \text{ is a positive integer}\}$$

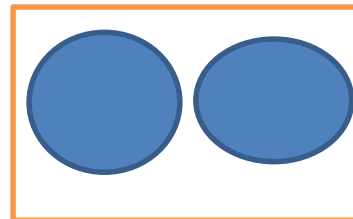
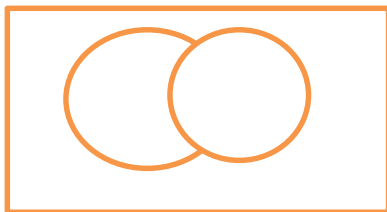
$$A = \{x | x \text{ is a positive even integer}\}$$

$$B = \{x | x \text{ is a positive even integer smaller than 10}\}$$

Then

$$A \cap B = \{x | x \text{ is a positive even integer smaller than 10}\}$$

If two set have no common element then they are said to be disjoint, or mutually exclusive. ,



Disjoint sets

Union:

The union of two sets A and B is the set of all elements belonging to A or B or both.

Symbolically denoted by $A \cup B$

Complementation:

The complement $\bar{A}$ of a set A is defined to be a set that contains all outcomes in the universal set U which do not belong to A.

Symbolically,

$$\bar{A} = \{x | x \in U, x \notin A\}$$

Note: A and $\bar{A}$ are mutually exclusive and their union is universal set.

Class of sets

A set whose members are sets themselves is called a class or family of sets.

The set $A = \{\{a\}, \{b\}, \{c\}, \{d, e\}\}$ is a family or class of sets.

Power set

The class of all subsets of a set A is called power set of A and is denoted by $\emptyset(A)$.

If $A = \{1, 2, 3\}$, then the power set of A is

$$\emptyset(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

The general rule is that if A is finite and has n elements, the power set will consist of 2^n elements including the null set.

Cartesian product set

Let A and B be two sets. The product set of A and B, denoted by $A \times B$, contains all ordered pairs (x, y) , where $x \in A$, $y \in B$.

Example: let $A = \{1, 2, 3\}$ and $B = \{4, 5\}$. Then the product set of A and B is

$$A \times B = \{(1, 4), (1, 5), (2, 4), (2, 5), (3, 4), (3, 5)\}$$

Difference of sets

The difference of sets A and B denoted by $A \setminus B$, is the set of elements which belongs to A but not belongs to B. Symbolically, $A \setminus B = \{x | x \in A, x \notin B\}$.

Algebra of sets

Cumulative laws:

1. $A \cup B = B \cup A$
2. $A \cap B = B \cap A$

Associative laws:

1. $(A \cup B) \cup C = A \cup (B \cup C)$
2. $(A \cap B) \cap C = A \cap (B \cap C)$

Distributive laws:

1. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
2. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Identity laws:

1. $A \cup \emptyset = A$
2. $A \cap U = A$

Idempotent laws:

1. $A \cup A = A$
2. $A \cap A = A$

Complement laws:

1. $A \cap \bar{A} = \emptyset$
2. $A \cup \bar{A} = U$

De Morgan's laws:

1. $\overline{(A \cup B)} = \bar{A} \cap \bar{B}$
2. $\overline{(A \cap B)} = \bar{A} \cup \bar{B}$

Experiment

A random experiment is a process with well-defined possible outcomes, but whose specific outcome cannot be predicted with certainty in advance. It's a fundamental concept in probability theory, used to model situations where chance plays a role.

Event

In probability, a random event is an occurrence with an unpredictable outcome. It's characterized by uncertainty, meaning you can't know the exact result beforehand.

Examples include flipping a coin or rolling a die, where the outcome (heads/tails, or a specific number) is not predetermined. While unpredictable, the likelihood of different outcomes can be analyzed using probability.

Sample space

A sample space of an experiment is the set of all possible outcomes of the same random experiment.

Sample Space

A **sample space** is the set of all possible outcomes in an experiment.

Example:
Two coins are tossed. Represent the sample space for this experiment by making a list, a table, and a tree diagram.
(H – Head, T – Tail)

List:

HH HT TH TT

Table:

	H	T
H	HH	HT
T	TH	TT

Tree Diagram:

HH
HT
TH
TT

The sample space is {HH, HT, TH, TT}

Sample point

Each possible outcome of a random experiment is called sample point or element in the sample space.

$S = \{HH, HT, TH, TT\}$ is a sample space where HH, HT, TH and TT are sample points.

Simple event

A simple event in probability is an event that has only one possible outcome.

Compound event

A compound event in probability involves the occurrence of two or more simple events.

Sure event and impossible event

a sure event is an event that is certain to happen, with a probability of 1. An impossible event is an event that cannot happen, with a probability of 0.

Sure Event:

- A sure event is an event that is guaranteed to occur when an experiment is performed.
- It encompasses the entire sample space of the experiment.
- The probability of a sure event is always 1.
- Example: Rolling a standard six-sided die and getting a number between 1 and 6.

Impossible Event:

- An impossible event is an event that has no chance of occurring.
- It is an event that is not possible to happen in a given experiment.
- The probability of an impossible event is always 0.
- Example: Rolling a standard six-sided die and getting a number greater than 6.

Mutually exclusive events

Events that cannot both happen at the same time.

Two or more events are mutually exclusive if the occurrence of one event prevents the occurrence of the other event(s).

Example:

In a single coin flip, the event of getting heads and the event of getting tails are mutually exclusive.

Probability:

If events A and B are mutually exclusive, the probability of either A or B occurring is the sum of their individual probabilities: $P(A \cup B) = P(A) + P(B)$.

Exhaustive events

Exhaustive events are a set of events where at least one event from the set must occur when an experiment is performed.

If both $A \cup B = S$ and $A \cap B = \emptyset$ are true then A and B are exhaustive events.

Example:

Rolling a die: The events "rolling a 1", "rolling a 2", "rolling a 3", "rolling a 4", "rolling a 5", and "rolling a 6" are exhaustive because they cover all possible outcomes.

Complementary events

An event A the event $\bar{A}$ consisting of all points of the sample space not in A, are called complementary events.

If A and $\bar{A}$ are complementary events then $A \cup \bar{A} = S$ and $A \cap \bar{A} = \emptyset$.

Counting Rules

Multiplication rule:

If an experiment can be performed in n_1 ways and if each of these, a second operation can be performed in n_2 ways and each of the first two operations, a third operation can be performed in n_3 ways, and so forth, then the number of ways in which the sequence of k operations can be performed is given by

$$n_1 \times n_2 \times n_3 \times \dots \times n_k$$

Furthermore, if the number of ways in which each of the k operations can be performed remains the same, say n , then the total number of ways would be

$$n \times n \times n \times \dots \times n = n^k$$

Example: Determine the number of sample points in the sample space when a pair of die is rolled.

Solution: The first die can land in six ways. For each of these six ways, the second die also can land in six ways. Therefore the number of ways in which the pair of dice can land is $6 \times 6 = 36$ or

$$6^2 = 36$$

Example: Suppose that there are three different routes from city A to city B and five different routes from city B to city C. then the number of different routes from city A to city C that pass through B is $3 \times 5 = 15$.

Permutation:

A permutation is an arrangement of objects in a specific order. It represents the different ways to arrange a set of items where the sequence matters.

In a permutation, the order in which the objects are arranged is crucial. Swapping the position of two objects creates a different permutation.

Examples:

- Arranging letters in a word: "CAT" is a different permutation than "ACT".
- Selecting a president, vice-president, and treasurer from a group of people: The order in which they are selected matters because it determines their roles.

The number of permutations of n objects taken r at a time is calculated using the formula:

$${}_nP_r = n! / (n - r)!$$

Example: How many permutations of four letters a, b, c and d can be made taking two at a time?

Solution: By the multiplication rule, we have two positions to fill with 4 choices for the first and 3 choices for the second for a total of $4 \times 3 = 12$ permutations. When enumerated, possible arrangements are: ab, ac, ad, ba, bc, bd, ca, cb, cd, da, db and dc. Here $n = 4$, and $r = 2$. Hence applying the general rule, the number of permutations of n distinct objects taken 2 at a time is

$${}_4P_2 = \frac{4!}{(4-2)!} = \frac{4 \times 3 \times 2 \times 1}{2 \times 1} = 12.$$

Combination:

A combination refers to the selection of items from a set where the order of selection does not matter.

Permutations vs. Combinations:

Permutations are distinct from combinations, where the order of selection does not matter. For example, choosing 2 fruits from a basket of apples, oranges, and bananas would be a combination, as the order of selection (e.g., apple then orange is the same as orange then apple) is not considered.

The number of permutations of n objects taken r at a time is calculated using the formula:

$${}^nP_r = \frac{n!}{r!(n-r)!}$$

Example: There are 20 people in a room of whom 12 are men and 8 are women. A committee of 3 is to be formed from them. How many ways can this be done? If it is desired that the committee will consist of 2 men and 1 woman, in how many ways can this be done?

Solution: It is evident that the order of the selected people is of no importance. Therefore, this is a problem of combination with $n = 20$ and $r = 3$. Thus

$${}^20C_3 = \frac{20!}{3!(20-3)!} = \frac{20!}{3!17!} = \frac{20 \times 19 \times 18 \times 17!}{3!17!} = 1140.$$

The number ways in which 2 men and 1 woman will be in the committee is

$${}^{12}C_2 \times {}^8C_1 = 66 \times 8 = 528.$$

Assigning probabilities

Basic steps:

1. Identifying the sample space.
2. Defining the event and
3. Assigning probabilities to the event.

Classical approach:

If an experiment has a total of $n(S)$ possible outcomes, all of which are mutually exclusive and equally likely, such that $n(A)$ of the outcomes are favorable to an event A , then the probability of the event A is defined by $P(A) = \frac{n(A)}{n(S)}$.

Example: A bag contains 4 white and 6 red balls. A ball is drawn at random from the bag. What is the probability that it is red? White?

Solution: A possible sample space for this experiment is $S = \{w, r\}$, where w stands for white and r stands for red balls. Then $n(S) = 10$. Let A is a set of white balls and B is a set for red balls. That means $n(A) = 4$ and $n(B) = 6$.

Hence $P(A) = \frac{n(A)}{n(S)} = \frac{4}{10} = 0.4$ and $P(B) = \frac{n(B)}{n(S)} = \frac{6}{10} = 0.6$

Relative frequency approach:

If an experiment repeated n times under essentially the same conditions when an event A occurs m times, then the limit of the function m/n tends to an idealized value p as n becomes infinitely large. This value is called the probability of the event A . symbolically,

$$p = P(A) = \lim_{n \rightarrow \infty} \frac{m}{n}$$

Subjective approach:

Subjective probability is the probability that an individual assigns to an event on the basis of his/her own experience, judgment, concept, intuition, information and beliefs.

Axiomatic approach:

Given the sample space S and the family of events in S , a probability function associated with each event A is real number $P(A)$, the probability of event A , such that the following axioms are satisfied:

Axiom 1: For any event A , $P(A) \geq 0$.

The axiom is referred as the axiom of positiveness.

Axiom 2: $P(S) = 1$.

This axiom referred as axiom of certainty.

Axiom 3: If A and B are two mutually exclusive events,

$$P(A \cup B) = P(A) + P(B)$$

This axiom is known as axiom of additivity.

Axiom 4: If $A_1, A_2, A_3, \dots, A_k$ is a sequence of k mutually exclusive events, then

$$P(A_1 \cup A_2 \cup A_3 \cup \dots \cup A_k) = P(A_1) + P(A_2) + P(A_3) + \dots + P(A_k)$$

Theorems on probability

Theorem 1: For any event A , $P(\bar{A}) = 1 - P(A)$.

This theorem states that if $\bar{A}$ is the complement of an event A , $P(\bar{A}) + P(A) = 1$.

Proof: Since A and $\bar{A}$ are disjoint events and $A \cup \bar{A} = S$,

$$P(A \cup \bar{A}) = P(S)$$

By axiom 2, $P(S) = 1$

By axiom 3, $P(A \cup \bar{A}) = P(A) + P(\bar{A})$

Therefore, $P(\bar{A}) + P(A) = 1$

Hence, $P(\bar{A}) = 1 - P(A)$.

Theorem 2: If $\emptyset$ is an empty set, $P(\emptyset) = 0$.

Proof: Since,

$$S \cup \bar{S} = S, S \cap \bar{S} = \emptyset \text{ and } \emptyset = \bar{S},$$

We have $P(\bar{S}) + P(S) = 1$ by axiom 3

Hence

$$\begin{aligned}
P(\emptyset) &= P(\bar{S}) \\
P(\emptyset) &= 1 - P(S) \\
P(\emptyset) &= 1 - 1 \\
P(\emptyset) &= 0
\end{aligned}$$

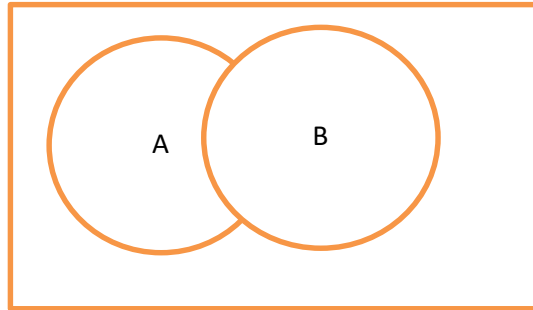
Theorem 3: $P(A) \leq 1$.

Proof: Since by axiom 1, $P(A) \geq 0$,
And by axiom 3, $P(\bar{A}) + P(A) = 1$,
It follows that, $P(A) \leq 1$.

Theorem 4: If $A \cap B \neq \emptyset$, then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Note: This theorem is variously known as the addition theorem, additive law or law of total probability.

Proof:



Look at the following Venn diagram. The union of A and B can be partitioned as follows:

$$A \cup B = (A \cap \bar{B}) \cup B$$

Therefore, we have $P(A \cup B) = P((A \cap \bar{B}) \cup B)$

$$\begin{aligned}
\text{Thus by axiom 3, } P(A \cup B) &= P(A \cap \bar{B}) + P(B) \\
&= P(A) - P(A \cap B) + P(B) \\
&= P(A) + P(B) - P(A \cap B)
\end{aligned}$$

Hence the proof.

Theorem 5: For any three events, A, B and C ,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C).$$

Example: Two coins are tossed. A is the event getting two heads and B is the event second coin shows head. Evaluate $P(A \cup B)$.

Solution: The sample space for this experiment is

$$S = \{HH, HT, TH, TT\}$$

And events A, B and $A \cap B$ are

$$A = \{HH\}, B = \{HH, TH\}, A \cap B = \{HH\}$$

And associated probabilities are

$$P(A) = \frac{1}{4}, P(B) = \frac{2}{4} \text{ and } P(A \cap B) = \frac{1}{4}$$

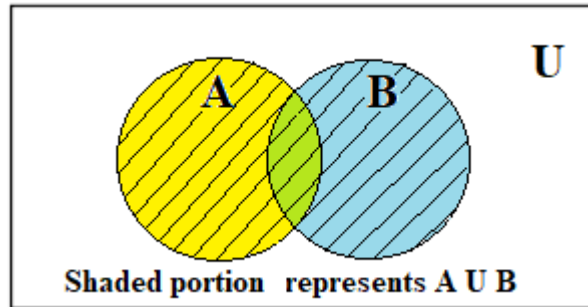
And hence

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\begin{aligned}
&= \frac{1}{4} + \frac{2}{4} - \frac{1}{4} \\
&= \frac{1}{2}
\end{aligned}$$

Theorem 6: For two events A and B, prove that $P[(\bar{A} \cap B) \cup (A \cap \bar{B})] = P(A \cup B) - P(A \cap B)$.

Proof: Look at the Venn diagram. Since the events $\bar{A} \cap B$ and $A \cap \bar{B}$ are mutually exclusive.



$$\begin{aligned}
P[(\bar{A} \cap B) \cup (A \cap \bar{B})] &= P(\bar{A} \cap B) + P(A \cap \bar{B}) \\
&= P(A) - P(A \cap B) + P(B) - P(A \cap B) \\
&= P(A) + P(B) - 2P(A \cap B) \\
&= P(A \cup B) - P(A \cap B)
\end{aligned}$$

Hence the proof.

Independence of events:

If A and B represents two events and if the occurrence of A does not affect, and is not affected by the occurrence of b, then A and B are said to be independent. In other words, two events are said to be independent if and only if $P(A \cap B) = P(A) \times P(B)$.

Example: A town has to fire engines operating independently. The probability that a specific fire engine is available when needed is 0.99.

- What is the probability that neither is available when needed?
- What is the probability that an engine is available when needed?

Solution: Let A be the event that the first engine is available when needed and B be the event that the second engine is available when needed. Then $P(A) = P(B) = 0.99$. Given this, the probability that both of them will be available when needed is $P(A \cap B) = P(A) \times P(B) = 0.99 \times 0.99 = 0.9801$, since they operate independently.

- Now $P(\bar{A} \cap \bar{B})$ is the probability that neither of them will be available when needed. Hence,

$$\begin{aligned}
P(\bar{A} \cap \bar{B}) &= P(\bar{A}) \times P(\bar{B}) \\
P(\bar{A} \cap \bar{B}) &= [1 - P(A)] \times [1 - P(B)] \\
&= 0.01 \times 0.01 \\
&= 0.0001
\end{aligned}$$

- $P(\text{an engine is available}) = P(A \cup B)$

$$= P(A) + P(B) - P(A \cap B)$$

$$\begin{aligned}
&= 0.99 + 0.99 - 0.9801 \\
&= 0.9999
\end{aligned}$$

Marginal probability:

Marginal probability is the probability of a single event occurring, regardless of the outcome of other events. It's essentially the probability of an event happening on its own, without considering any other factors or conditions.

Example: Two hundred students were classified according to their faculty they belong to and their gender. The distribution is as follows:

Faculty	Gender		Total
	Male(M)	Female(F)	
Commerce(C)	36	24	60
Humanities(H)	84	56	140
Total	120	80	200

From the table,

$n(M) = 120$, $n(F) = 80$, $n(C) = 60$, $n(H) = 140$ and $n(S) = 200$.

The probability of selecting a male student denoted by $P(M)$ is

$$P(M) = \frac{n(M)}{n(S)} = \frac{120}{200} = 0.60$$

And that of selecting a female student, denoted by $P(F)$ is

$$P(F) = \frac{n(F)}{n(S)} = \frac{80}{200} = 0.40$$

Similarly, probability of selecting a commerce student is

$$P(C) = \frac{n(C)}{n(S)} = \frac{60}{200} = 0.30$$

And that of selecting a humanities student is

$$P(H) = \frac{n(H)}{n(S)} = \frac{140}{200} = 0.70$$

Joint probability:

Joint probability refers to the probability of two or more events occurring simultaneously. It represents the likelihood that all of these events will happen at the same time. Symbolically,

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)}$$

From the table above, probabilities of the joint events are

$$P(F \cap C) = \frac{24}{200}, P(M \cap H) = \frac{84}{200}$$

The joint probabilities of the event can be put in the tabular form:

Faculty	Gender		Total
	Male(M)	Female(F)	
Commerce(C)	0.18	0.12	0.30
Humanities(H)	0.42	0.28	0.70
Total	0.60	0.40	1.00

Theorem: If the events A and B are independent, and $\bar{A}$ and $\bar{B}$ are respectively their complementary events, then the following pairs of events are also independent:

- (i) A and $\bar{B}$ (ii) $\bar{A}$ and B (iii) $\bar{A}$ and $\bar{B}$

Proof: Since A and B are independent, $P(A \cap B) = P(A)P(B)$

Now,

$$\begin{aligned}
 P(A \cap \bar{B}) &= P(A) - P(A \cap B) \\
 &= P(A) - P(A)P(B) \\
 &= P(A)[1 - P(B)] \\
 &= P(A)P(\bar{B})
 \end{aligned}$$

Hence A and $\bar{B}$ are independent.

$$\begin{aligned}
 P(\bar{A} \cap B) &= P(B) - P(A \cap B) \\
 &= P(B) - P(A)P(B) \\
 &= P(B)[1 - P(A)] \\
 &= P(\bar{A})P(B)
 \end{aligned}$$

This shows that $\bar{A}$ and B are independent.

To prove that $\bar{A}$ and $\bar{B}$ are independent, we find that

$$\begin{aligned}
 P(\bar{A} \cap \bar{B}) &= P(\bar{A}) - P(\bar{A} \cap B) \\
 &= P(\bar{A}) - P(\bar{A})P(B) \text{ ;(since } \bar{A} \text{ and B are independent)} \\
 &= P(\bar{A})[1 - P(B)] \\
 &= P(\bar{A})P(\bar{B})
 \end{aligned}$$

Hence $\bar{A}$ and $\bar{B}$ are independent.